William Chuang **Curriculum Vitae**

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Education

University of Arizona, Tucson, AZ

M.S. in Mathematics May 2025

Originally admitted to the Ph.D. program in Mathematics; completed the M.S. following a formal program transition.

University of Arizona

M.S. in Mathematics Spring 2025

Graduate-level coursework completed under Ph.D. classification, with formal study concentrated in core analysis (measure theory, functional analysis, Fourier analysis), algebra, geometry, and topology.

In parallel, pursued independent, self-directed research *outside of formal university courses or sponsored projects* in cryptographic theory, hyperbolic geometry, and deep learning architecture. This work focused on the interface of symbolic dynamics, hyperbolic geometry, cryptographic architecture, and neural representation theory, including the GRAIL meta-architecture for geometry-native neural computation and the Modular Spectral Indistinguishability Assumption (MSIA) for post-quantum cryptography; portions were submitted to DARPA-SN-25-60 and disclosed in a U.S. provisional patent application (filed March 2025).

Graduate Teaching Assistant (College Algebra, Calculus I), 2022–2025; Tucson Math Circle volunteer webmaster; RTG seminar participant.

San Francisco State University, San Francisco, CA

M.A. in Mathematics May 2022

Thesis: *The Hausdorff Dimension of Limit Sets of Well-Distributed Schottky Groups*

Link: `https://scholarworks.calstate.edu/downloads/xd07h079g`

Advisor: Dr. Chun-Kit Lai

Coursework emphasis in real analysis, measure theory, functional analysis, algebra, and topology.

Graduate Instructor for Precalculus (25 students) and Teaching Assistant for Calculus I–II (125+ students); led problem sessions and labs, created original problem sets, and held regular office hours.

Research projects on transformer decomposition via spectral cycles, entropy-critical attention mechanisms, and modular-trace scoring functions for cryptographic verification, with reproducible PyTorch implementations.

University of San Francisco, San Francisco, CA

B.S. in Mathematics (Honors Major), Minor in Computer Science Dec 2018

Major GPA: 3.88/4.00 **Honors:** Cum Laude

Dean's Honor Roll (multiple terms); member of Pi Mu Epsilon (U.S. Mathematics Honor Society) and President's Ambassador.

Independent number theory research and early work in symbolic and statistical foundations for machine learning; transcribed and edited lecture notes for MSAN 504 (now MSDS 504: Review of Probability and Statistics).

Pennsylvania State University, University Park, PA

Mathematics Advanced Study Semesters (MASS), Visiting Scholar 2017–2018

Competitive Ph.D.-preparatory program with full tuition waiver and internal scholarships.

Completed graduate-level tracks in Knot Theory, Functional Analysis, and Elliptic Functions & Curves, with additional seminars in topology, Lie theory, dynamical systems, and mathematical physics.

National Taiwan University, Taipei, Taiwan

Undergraduate studies in Theoretical and Mathematical Physics 2010–2014

Transferred from National Dong Hwa University into Bioenvironmental Systems Engineering while independently pursuing advanced coursework and seminars in quantum field theory, gravitation, cosmology, and AdS/CFT at NTU and Academia Sinica (LeCosPA).

National Dong Hwa University, Hualien, Taiwan

Undergraduate studies in Physics 2007–2010

Class rank 1/55; four-time recipient of the Dean's Honor Prize.

Early research training in theoretical and experimental physics, including electrodynamics, relativity, gravitation, quantum mechanics, and thermodynamics, laying the foundation for later work in geometric learning and post-quantum cryptography.

Research Interests

I am broadly interested in the interplay between mathematical physics, geometry, topology, and the theory of intelligent systems. My current work focuses on building a unified, observer-centric framework that connects quantum theory, general relativity, geometry, and modern machine learning architectures. Representative directions include:

- **Observer-Centric Quantum Gravity and Dual Quantization (Lambda-Stack, GRAIL $\times$ DFA).** I am developing an observer-centric framework in which quantization arises from the non-commutative dynamics of the observer rather than by quantizing spacetime itself. The core architecture is a tri-layer "lambda-stack":
  - a *geometry* layer (automorphic kernels on hyperbolic or Lorentzian manifolds),
  - a *symbolic* layer (deterministic finite automata (DFA) that gate legal code paths),
  - a *thermodynamic* layer (a controlled entropy-and-scheduling mechanism, CEAS).

  Together with the GRAIL "optimizer-as-connection" viewpoint (curvature from commutators of symmetry flows with gradient flows), this yields a non-commutative effective theory (Schrödinger/Fokker–Planck, Langevin on the Fisher manifold, retarded/heat kernels, KMS-like relations) that is internally quantum-like for the observer and compatible with standard QM/QFT/GR when treated as an inference-level theory.

- **Metric-Invariant Architecture (MIA) and Geometric Compute.** I study architectures that replace standard scalar multiplication/division by *metric-invariant* primitives $F \circ I$ on a manifold (M, g), where I is a group-invariant scalar (distance, bilinear form, cross-ratio, Casimir, etc.) and F is a calibrated head map that reproduces IEEE-754 behavior. At the system level this supports:
  - bit-exact or last-bit-safe replacement of floating-point multiply,
  - orbit-locked model twins (function-identical, internally distinct deployments),
  - sovereign compute stacks on mature nodes and heterogeneous substrates (digital, analog, and in-/near-memory),
  - architecture-level obfuscation and security-by-geometry.
- **Quantum-Ready Transformer/Autoencoder Architectures from Geometric First Principles.** I design transformer and autoencoder architectures in which attention and encoding dynamics arise from Möbius flows in $\mathrm{PSL}(2, \mathbb{R})$, generated by differential equations of the form

$$\frac{dz}{dt} = \kappa\,(z - z_-)(z - z_+),$$

  corresponding to elements of $\mathfrak{sl}(2, \mathbb{R})$. Each layer is constructed as the exponential of a Lie-algebra generator, yielding a one-parameter flow that preserves hyperbolic geometry. This framework:
  - satisfies the usual formal criteria of a quantum system (unitary evolution, Hermitian generators, complex Hilbert space, Lie algebraic structure, discrete spectral dynamics),
  - supports exact quantization via Schrödinger-type evolution under unitary group representations,
  - encodes model evolution as geodesic motion on curved latent manifolds, improving interpretability and geometric structure preservation,
  - provides a hardware-compatible foundation for quantum large language models and quantum-ready encoders.
- **Automorphic Attention, Hyperbolic Geometry, and Kleinian Groups in Neural Networks.** I develop attention mechanisms and representation layers based on automorphic forms (for example, Maass–Poincaré series and Hecke eigenfunctions) on hyperbolic spaces and their quotients. This includes:
  - automorphic and modular-invariant attention kernels,
  - group actions (Fuchsian/Kleinian groups) as global symmetry operators on latent space,
  - hyperbolic and pseudo-Riemannian latent geometries for stability, expressivity, and security.

I view these constructions as geometric analogues of monodromy: global information about the network is encoded by how states transform under loops in parameter space and under group actions on latent manifolds.

- **Monodromy, Langlands, and Category-Theoretic Structures in Training Dynamics.** I study the global evolution of neural networks as parameters move along loops (monodromy), and relate this to deeper structures from the Langlands program and representation theory. This includes:
  - monodromy of training trajectories as a topological invariant of optimization,
  - categorical correspondences between automorphic representations and Galois representations used as meta-training signals,
  - functorial update rules that attempt to predict or shortcut gradient-based training by algebraic means.
- **Geometric and Topological Foundations of Self-Attention and Deep Architectures.** I investigate intrinsic factors and topological structures underlying self-attention architectures and deep networks more broadly:
  - defining abstract notions of curvature and connection on spaces of models and data,
  - studying linear combinations, cyclic structures, and orbifolds of feedforward networks with restricted codomains,
  - using autoencoders and their inverses as tools for model interpretability and controlled editing.
- **Renormalization, Motives, Hopf Algebras, and Information Geometry in Deep Learning.** I aim to build a rigorous interaction theory for deep neural networks by integrating:
  - Callan–Symanzik-type equations and stochastic measures,
  - renormalization group structures encoded via Hopf algebras and motives,
  - random matrix theory, Fisher information geometry, and statistical field theory,

  to better understand scaling laws, phase transitions, and universality phenomena in modern deep learning.
- **Vacuum Thermodynamics, Strong-Gradient Control, and Vacuum-Aging Engines.** Motivated by semiclassical gravity and condensed-matter analogues, I explore theoretical and experimental roadmaps for:
  - dual-resonant "slingshot" control stacks (mechanical plus electromagnetic) that sculpt strong, localized stress-energy gradients,
  - vacuum-centric thermodynamic models in which finite regions of spacetime are treated as non-maximal-entropy ensembles,
  - controllable vacuum-aging engines and analogue systems that probe Schwinger-like or Lee–Yang-type critical behavior in laboratory settings.

Patent Applications and Intellectual Property

All filings below are U.S. patent applications currently in provisional or non-provisional status; none have yet been examined to issuance.

- **Metric-Invariant Architecture**
  US Provisional Patent Application 63/901,369 — Filed October 17, 2025
  Establishes a metric-invariant architecture for learning systems, providing a general framework whose behavior is stable under changes of underlying geometric metric. Filed under 35 U.S.C. §111(b) to secure an early priority date.
- **Operator-Theoretic Verification of Neural Models**
  US Provisional Patent Application 63/881,657 — Filed September 15, 2025
  Introduces an operator-theoretic framework for interpretable and certifiable neural networks (including autoencoders, transformers, and large language models), enabling audit-grade guarantees on coverage and safety, as well as projector-level edits to model behavior without retraining.
- **Analytic Corridor Methods for Critical Synchronization, Memory Integration, and Optimization in Ising-like Neural Systems with Applications to Mind Upload, Modular Restoration, and AI Global Optimization**
  US Provisional Patent Application 63/824,102 — Filed June 15, 2025
  Develops analytic "corridor" methods for critical synchronization and memory integration in Ising-like neural systems, with applications to coordinated AI memory, modular restoration, and global optimization. The work is theoretical and unclassified.
- **System and Method for Remote Polygraphy and Internal Voice Capture for Neural Training via Multimodal Electromagnetic Sensing**
  US Provisional Patent Application 63/824,053 — Filed June 15, 2025
  Proposes an experimental, privacy-conscious sensing concept for multimodal signal acquisition to support AI model training. The design is exploratory, unclassified research and has not been deployed.
- **Neural Projection Unit with CEAS-Driven Dynamic Beta Scaling and Thermodynamic Synchronization**
  US Patent Application 19/232,791 — Filed June 9, 2025
  Describes an optimization module for neural networks that dynamically scales a key parameter (β) according to critical-entropy attention system (CEAS) principles, improving stability and convergence through thermodynamic synchronization.
- **Neural Projection Unit with CEAS-Based Dynamic Beta Scaling and Thermodynamic Synchronization**
  US Provisional Patent Application 63/820,652 — Filed June 9, 2025
  Companion provisional application covering related design space and parameter-scaling variants for the CEAS-based neural projection unit, supporting subsequent non-provisional filings and academic research.
- **Critical Entropy Attention System (CEAS)**
  US Provisional Patent Application 63/813,617 — Filed May 29, 2025

Explores an attention mechanism that uses critical-entropy principles to optimize energy and computation in neural architectures; unclassified, academic R&D with potential applications to robust and efficient AI systems.

- **The Modular Symbolic Intelligence Architecture (MSIA)**
  US Provisional Patent Application 63/809,257 — Filed May 20, 2025
  Defines a modular framework for symbolic–neural hybrid AI systems, integrating symbolic dynamics with neural representations to improve interpretability, robustness, and long-horizon reasoning. Unclassified and exploratory.
- **Hyperbolic Autoencoder and Transformer Framework for Real-Time Landscape Modification and Spacetime Adaptation**
  US Provisional Patent Application 63/773,441 — Filed March 18, 2025
  Presents a geometric deep-learning framework using hyperbolic latent spaces for real-time adaptation of optimization landscapes, with a working prototype in academic research settings.

Original Theoretical Frameworks and (Conjectured) Historical Firsts

The following original contributions each satisfy the formal requirements of a mathematical theory: *explicitly defined objects, internal axioms, morphisms (or dynamics), and closure under composition.* They also represent, to the best of my knowledge and based on available literature and records, historically first frameworks or combinations of geometry, physics, and machine learning.

They span foundational neural algebra, quantum dynamics, symbolic cryptography, arithmetic representation theory, and observer–centric effective field theory. Several are interconnected and modular, forming a self–consistent architecture of novel post–quantum and intelligence–relevant learning systems.

(1) **GRAIL (Geometric Representation Algebra for Intelligent Learning): Geometry–Native Neural Meta–Architecture and Hyperbolic Neural Algebra.**

- *Theory.* Encodes neural computation as algebraic operations over curved manifolds (hyperbolic, Lorentzian, modular, etc.), specified by symmetry groups, Lie algebras, and categorical morphisms, rather than a priori Euclidean coordinates. Representation–theoretic, inner–product implementations form only a restricted subclass: GRAIL admits infinitely many operator families, including geometric, symbolic, entropic, and cryptographic constructions.
- Provides a fully geometry–consistent, hyperbolically grounded framework in which:
  - representations live in hyperbolic or Kleinian quotients;
  - group actions provide global symmetries and orbit mobility over models;
  - the algebra of activations and weights is reinterpreted through curved geometry rather than purely Euclidean operations.
- **Training regimes:**
  - Backpropagation–compatible curved–space operators on Riemannian or pseudo–Riemannian manifolds.

– Symbolic, non–differentiable operators (e.g., Langlands layers, cryptographic kernels, DFA lifts) trained via monodromy, Lorentzian landscape reshaping, fixed–point iteration, or categorical dynamics.
- *Historical aspect.* To the best of my knowledge, this is the first unified meta–architecture that treats neural computation over general curved spaces via group–theoretic and categorical structure, subsuming hyperbolic neural networks, Lorentzian models, and geometry–native cryptographic and symbolic components.

(2) **Metric–Invariant Architecture (MIA): Invariant–First Arithmetic, Twin Models, and Laptop–Reproducible Proof–of–Work.**

- *Theory.* Replaces scalar multiplication and division with metric–invariant primitives of the form

$$\widehat{\mu}(x,y) \;=\; \iota\big(F(I(x,y))\big), \qquad \pi\big(\widehat{\mu}(\iota(a),\iota(b))\big) = \mathrm{round}_{754}(a\times b),$$

where Σ is the floating–point domain, $\iota:\Sigma\to M$ and $\pi:M\to\Sigma$ are encoding/decoding maps on a manifold (M,g) with $\pi\circ\iota=\mathrm{id}_\Sigma$, $I(x,y)$ is a group–invariant scalar (distance, bilinear form, cross–ratio, etc.), and F is a calibrated readout reproducing IEEE–754 multiplication (including subnormals, signed zeros, infinities, and NaNs, up to last–bit corrections).
- *Twinhood and architecture.* Uses diagonal transport (apply the same group action to inputs, internal state, and encoded outputs) to construct infinite families of *twin* models: function–identical but internally distinct realizations that support security, obfuscation, and per–tenant deployment without exposing plaintext parameters. Admits a verification blueprint with:
  – bit–exact fp16 primitives ($F\circ I$) tested exhaustively against IEEE–754 multiplication (values and flags);
  – whole–machine diagonal transport laws;
  – game–based security notions (twin–indistinguishability and orbit–recovery resistance);
  – proof scaffolding (Coq/Isabelle plus SMT harnesses) and CI gates (bit–match, invariance tests, trace checks).
- *Validation and proof–of–work.* Provides a validation pipeline that:
  – calibrates peak dot–product throughput and bandwidth on commodity hardware;
  – places MIA kernels on a calibrated roofline chart using measured arithmetic intensity and achieved Gops/s;
  – demonstrates that, for certain workloads, MIA kernels operate within compute and bandwidth ceilings without requiring sub–100 nm silicon.
- *Historical aspect.* To the best of my knowledge, this is the first explicit, IEEE–754–true, metric–invariant arithmetic layer designed for conventional machines, equipped with a full verification story and laptop–reproducible roofline proof–of–work.

(3) **Lambda–Stack Dual Quantization (Observer–Centric Dual Quantization): Triadic Quantization via Geometry, DFA, and Thermodynamics.**

- *Theory.* Constructs a tri–quantized observer architecture with three coupled pillars:
  (a) A geometry layer using automorphic kernels on curved spaces (e.g., hyperbolic attention on $\mathbb{H}^2$ or higher).
  (b) A symbolic layer given by deterministic finite automata (DFA) and codebook dynamics, in which cycle components lift to unitary blocks and transients lift to completely positive, trace–preserving (CPTP) channels.
  (c) A thermodynamic layer (CEAS) regulating an effective inverse temperature β and enforcing critical–entropy corridors.
- Together with the GRAIL "optimizer as connection" view (curvature from commutators of symmetry flows with gradient flows), this yields a non–commutative operator algebra for the observer: Lie brackets measured via Baker–Campbell–Hausdorff drift, unitary cycle blocks, CPTP transient channels, and an effective theory with Schrödinger/Fokker–Planck duality, Langevin dynamics on the Fisher manifold, retarded/heat kernels, and random–matrix descriptions of model twins.
- The key stance is to *quantize the observer* (its algebra of observables and update rules) rather than directly quantizing the metric, while remaining compatible with QM/QFT/GR when interpreted as a meta–theory of inference.
- *Historical aspect.* To the best of my knowledge, this is the first explicit dual–quantization programme for an observer–centric physics engine built on a triad of automorphic geometry, DFA quantization, and thermodynamic control.

(4) **First–Principle Quantum Neural Dynamics (FPQND) and Quantum–Ready Transformer Architectures.**
- *Theory.* Formulates neural dynamics under Schrödinger–type equations derived from Lie group symmetries and Hamiltonian flows. Network evolution is described by Hermitian generators and unitary propagators on complex Hilbert spaces, with discrete spectral dynamics corresponding to layerwise updates.
- *Quantum–ready transformers.* Designs transformer and autoencoder architectures in which each layer is realized as the exponential of a generator in $\mathfrak{sl}(2,\mathbb{R})$, corresponding to a Möbius transformation in $\mathrm{PSL}(2,\mathbb{R})$. The architecture:
  – is derived from differential–geometric first principles;
  – satisfies standard quantum–system criteria (unitary evolution, Hermitian Hamiltonians, Hilbert space representation, discrete spectra);
  – encodes model evolution as geodesic motion on curved latent manifolds, enabling exact quantization and geometry–aware interpretability.
- *Historical aspect.* To the best of my knowledge, this is the first transformer/autoencoder architecture built from geometric first principles that rigorously satisfies all formal criteria of a quantum system and is intended as a physically valid basis for quantum neural computation.

(5) **Langlands–Neural Correspondence (LNC), Automorphic Attention, and Langlands–Driven Cryptographic Design.**

- *Theory (LNC).* Embeds Langlands duality into neural training dynamics by treating automorphic representations (modular forms, Maass eigenfunctions) and Galois representations (Frobenius eigenvalues, Satake parameters) as objects in categories $\mathcal{A}$ and $\mathcal{G}$, connected via a Langlands functor

$$\mathcal{L} : \mathcal{A} \to \mathcal{G}.$$

  Functorial update rules use arithmetic spectral data to replace or augment gradient descent, providing closed–form, symbolic predictions of weight updates and hyperparameter flows.
- *Automorphic attention (QLAI).* Implements LNC in attention layers via automorphic kernel activations (e.g., Maass–Poincaré series, Hecke eigenfunctions) and Hecke operator spectral filtering, yielding modular– and automorphic–invariant attention propagation and spectrally structured representations.
- *Langlands–based crypto.* Proposes transformer architectures in which structural analogues of Langlands reciprocity are embedded at the representation level: automorphic features act as "modular" objects, dual key structures play the role of Galois representations, and functorial correspondences determine obfuscation parameters and key schedules.
- *Historical aspect.* To the best of my knowledge, this is the first formal realization of Langlands correspondence in a machine–learning context, and the first use of automorphic forms as the mathematical basis of attention mechanisms and neural cryptographic design.

(6) **Monodromic Training Dynamics and Lorentzian Loss–Landscape Control.**
- *Monodromic dynamics.* Introduces monodromy and homotopy group actions to track how deep models evolve when parameters traverse loops in moduli space, extending the classical notion from Fuchsian differential equations to modern AI models. Training is viewed as a global, topological process; loop classes define invariants of the optimization landscape.
- *Lorentzian control.* Formulates a curvature–aware alternative to pure gradient descent in which Lorentz transformations (isometries of Minkowski space) are used to reparameterize weights and reshape loss landscapes, enabling:
  - loss tunneling along lightlike or nearly lightlike directions;
  - gradient teleportation via global reparameterizations;
  - global convergence strategies following relativistic geodesics.
- *Historical aspect.* To the best of my knowledge, this is the first theory that simultaneously (i) treats training as monodromy over parameter loops and (ii) uses Lorentzian geometry as a principled landscape control mechanism in machine learning optimization.

(7) **Curved–Space Neural Encryption (CNE), Holographic Bulk–Boundary Neural Encryption, and AdS/CFT–Inspired Cryptosystems.**
- *Curved–space duality theory.* Establishes a general cryptographic framework based on bulk–boundary dualities between deep neural "bulk" layers and latent "boundary"

representations in curved spaces defined by GRAIL. Encryption and decryption correspond to bulk–boundary transforms; partial inversion is equivalent to reconstructing bulk geometry from boundary data, yielding hardness analogues.

- *Holographic neural encryption.* Designs cryptographic neural architectures in which
  - input tokens act as boundary observables;
  - internal transformer layers behave as bulk fields on a curved latent manifold;
  - weight dynamics play the role of a renormalization–group flow;
  - decryption corresponds to bulk reconstruction from boundary signals.
- *AdS/CFT–inspired implementation.* Implements a concrete cryptosystem whose latent geometry is modeled on an AdS–like hyperbolic $(n + 1)$–dimensional manifold, with outputs interpreted as CFT–like boundary observables. The duality is encoded symbolically in the tensor and attention structure; it does not assume our universe is AdS.
- *Historical aspect.* To the best of my knowledge, this is the first family of neural cryptosystems that explicitly encode holographic and AdS/CFT–style bulk–boundary dualities in transformer architectures and use the intractability of bulk reconstruction as a cryptographic hardness source.

(8) **Modular Spectral Indistinguishability Assumption (MSIA): Post–Quantum Spectral Cryptography.**

- *Theory.* Introduces a cryptographic hardness framework based on symbolic dynamics, modular trace algebras, and factorization/inversion problems for structured zeta functions
$$\zeta_p(z) \; = \; \frac{1}{\det(I - zA^{(p)})}, \qquad A^{(p)} \in \mathbb{F}_p^{n\times n},$$
whose coefficients encode rich combinatorial and representation–theoretic constraints (e.g., projective indecomposable modules, symbolic Brauer classes).
- Supports trapdoor constructions, parameter amplification (including layered obfuscation) and is compatible with GRAIL, MIA, and CNE.
- *Historical aspect.* To the best of my knowledge, this is the first post–quantum cryptographic primitive built on modular trace zeta functions and symbolic module structures within a structured, spectral–theoretic framework.

(9) **Fractal Cryptographic Dynamics (FCD) and Topological Entropy as a Hardness Parameter.**

- *Theory.* Uses fractal dimension and symbolic entropy as tunable cryptographic hardness parameters. Data and keys are encoded into dynamically generated fractal sets and symbolic systems whose inversion is equivalent to reconstructing invariant sets from partial observations.
- Embeds Hausdorff dimension δ and topological entropy h_{top} as explicit levers that control complexity, obfuscation strength, and resistance to classical and quantum attacks.
- *Historical aspect.* To the best of my knowledge, this is the first explicit proposal to use fractal dimension and topological entropy as primary, tunable hardness parameters in cryptographic and neural obfuscation schemes.

(10) **Arithmetic–Neural Quantum Field Theory (ANQFT): Loop Integrals via Neural–Automorphic Objects.**

- *Theory.* Proposes a functorial mapping from all–loop–order Feynman integrals and quantum periods to neural–algebraic automorphic forms within the GRAIL framework, organizing scattering amplitudes and periods through neural moduli spaces and automorphic basis functions.
- *Historical aspect.* To the best of my knowledge, this is the first attempt to encode general quantum field–theoretic loop integrals and periods directly into neural/automorphic representations for symbolic simplification and structural analysis.

(11) **Cosmic Neural Galois Representation (CNGR): Arithmetic Symmetries in Learnable Systems.**

- *Theory.* Defines a category of neural–motivic objects encoding mixed motives, Galois actions, and period structures, designed to simulate a "cosmic" Galois group acting on learnable architectures. Constructs Frobenius–compatible update rules, L–function representations, and symbolic descent mechanisms that propagate arithmetic symmetries through neural parameters.
- *Applications.* Symbolic arithmetic learning, automorphic AI alignment, arithmetic period inference, and post–quantum cryptographic resilience via neural–motivic duality.
- *Historical aspect.* To the best of my knowledge, this is the first explicit proposal to realize a cosmic–Galois–style action inside learnable systems via neural–motivic categories.

(12) **Vacuum–Graviton Cascades, Lee–Yang Vacuum Engineering, and Quantum Amplification Cascades.**

- *Vacuum–aging engines and cascades (VGCF).* Formalizes vacuum bursts as curvature–amplified excitations, unifying Coulomb slingshots, nuclear amplification, and gravitational Schwinger mechanisms within a coherent cascade framework. Proposes *vacuum–aging engines* in which finite spacetime regions, viewed as non–maximal–entropy ensembles, relax and release usable free energy under strict energy accounting, using dual–resonant (mechanical plus electromagnetic) "slingshot" control stacks.
- *Dynamic Lee–Yang Vacuum Engineering (DLYVE).* Embeds Lee–Yang zeros into partition structures of condensed matter and analogue systems to tune critical lines and corridors for vacuum–like phase transitions, coupling finite–size scaling, susceptibilities, and Lee–Yang analysis to experimental control parameters.
- *Quantum Amplification Cascade Theory (QACT).* Builds an operator–algebraic and path–integral framework for vacuum amplification cascades, including induced stress–energy tensors and coherence conditions across multiple stages. Integrates traceless or nearly traceless vacuum burst stress–energy tensors (to leading order) as backreaction regulators, keeping gravitational response in the perturbative regime.
- *Historical aspect.* To the best of my knowledge, this is the first unified programme that (i) combines dual–resonant control, Lee–Yang criticality, and gravitational Schwinger–type amplification into a coherent laboratory vacuum–engineering framework, and (ii) treats the resulting cascades with explicit operator and path–integral structure.

Publications and Preprints (DOI-Indexed)

(1) William Huanshan Chuang, "Dual–Resonance Slingshot Control and Vacuum–Aging Engines," preprint, 2025. DOI: 10.5281/ZENODO.17527311.
(2) William Huanshan Chuang, "Observer–Centric Quantum Gravity as Dual Quantization," preprint, 2025. DOI: 10.5281/ZENODO.17508503.
(3) William Huanshan Chuang, "Metric-Invariant Architecture from GRAIL: A Verified Path to Group-Invariant Programs, Bit-Exact Floats, and Diagonal Transport of Machines," preprint, 2025. DOI: 10.5281/ZENODO.17401675.
(4) William Huanshan Chuang, "The Global Impact of Metric-Invariant Architecture (MIA)," preprint, 2025. DOI: 10.5281/ZENODO.17394706.
(5) William Huanshan Chuang, "Feasibility of Replacing Scalar Multiplication with Metric-Invariant Functions on Traditional Machines," preprint, 2025. DOI: 10.5281/ZENODO.17382332.
(6) William Huanshan Chuang, "Prime and Simple Geodesics on Hyperbolic Surfaces: A Survey," preprint, 2025. DOI: 10.5281/ZENODO.17363657.
(7) William Huanshan Chuang, "A Structural Approach via Central Characters and Graph Theoretic Methods," preprint, 2025. DOI: 10.5281/ZENODO.17362262.
(8) William Huanshan Chuang, "Central Characters, Blocks, Brauer Graphs, and Kernel Decompositions," preprint, 2025. DOI: 10.5281/ZENODO.17361874.
(9) William Huanshan Chuang, "Critical Scaling in Hyperbolic Attention Mechanisms," preprint, 2025. DOI: 10.5281/ZENODO.17363323.
(10) William Huanshan Chuang, "Decomposition Matrices, GAP Computations, and Frobenius–Schur Indicators," preprint, 2025. DOI: 10.5281/ZENODO.17354868.
(11) William Huanshan Chuang, "Extra Lecture – The Set $\mathrm{IBr}(G)$: Irreducible Brauer Characters," preprint, 2025. DOI: 10.5281/ZENODO.17361477.
(12) William Huanshan Chuang, "Extra Lecture – The Structure and Properties of $\mathrm{Irr}(G)$," preprint, 2025. DOI: 10.5281/ZENODO.17361545.
(13) William Huanshan Chuang, "Formal Verification and Cryptanalysis of AdS/CFT-based Hyperbolic Cryptographic System," preprint, 2025. DOI: 10.5281/ZENODO.17363492.
(14) William Huanshan Chuang, "Hausdorff Dimension of Well-Distributed Schottky Groups," preprint, 2025. DOI: 10.5281/ZENODO.17363252.
(15) William Huanshan Chuang, "Hyperbolic Orbit-Based Encryption via Neural Geodesic Cycles," preprint, 2025. DOI: 10.5281/ZENODO.17363387.
(16) William Huanshan Chuang, "Hyperbolic Transformers – Theory and Implementation," preprint, 2025. DOI: 10.5281/ZENODO.17363551.
(17) William Huanshan Chuang, "Hyperbolic Transformers – Theory and Implementation," preprint, 2025. DOI: 10.5281/ZENODO.17363522.
(18) William Huanshan Chuang, "Irreducibility Criteria, Frobenius–Schur Analysis, and Modular Representation Implications," preprint, 2025. DOI: 10.5281/ZENODO.17354912.

(19) William Huanshan Chuang, "Lecture 1 – Lifting Brauer Characters and Generalized Class Functions," preprint, 2025. DOI: 10.5281/ZENODO.17354022.
(20) William Huanshan Chuang, "Lecture 1 – Representations and Characters of Finite Groups," preprint, 2025. DOI: 10.5281/ZENODO.17354935.
(21) William Huanshan Chuang, "Lecture 10 – Examples: Symmetric and Linear Groups, Block Structures, and Invariants," preprint, 2025. DOI: 10.5281/ZENODO.17354791.
(22) William Huanshan Chuang, "Lecture 10 – Synthesis, Advanced Examples, and Research Directions in Modular Character Theory," preprint, 2025. DOI: 10.5281/ZENODO.17361436.
(23) William Huanshan Chuang, "Lecture 11 – Global Conjectures: Alperin, Dade, and Donovan," preprint, 2025. DOI: 10.5281/ZENODO.17354813.
(24) William Huanshan Chuang, "Lecture 12 – Course Wrap-Up and Modern Perspectives," preprint, 2025. DOI: 10.5281/ZENODO.17354828.
(25) William Huanshan Chuang, "Lecture 2 – Decomposition Matrices and Projective Indecomposable Characters," preprint, 2025. DOI: 10.5281/ZENODO.17354030.
(26) William Huanshan Chuang, "Lecture 2 – Modular Representations and Brauer Characters," preprint, 2025. DOI: 10.5281/ZENODO.17354959.
(27) William Huanshan Chuang, "Lecture 3 – Block Theory, Defect Groups, and Decomposition Matrix Structure," preprint, 2025. DOI: 10.5281/ZENODO.17354075.
(28) William Huanshan Chuang, "Lecture 3 – Irreducibility over Finite Fields and Frobenius Automorphisms," preprint, 2025. DOI: 10.5281/ZENODO.17355010.
(29) William Huanshan Chuang, "Lecture 4 Supplement – Brauer's Theorems, the Cartan Matrix, and Loewy Structure," preprint, 2025. DOI: 10.5281/ZENODO.17361786.
(30) William Huanshan Chuang, "Lecture 4 – Brauer Character Tables and Decomposition Matrices," preprint, 2025. DOI: 10.5281/ZENODO.17360738.
(31) William Huanshan Chuang, "Lecture 4 – Brauer's Theorems, the Cartan Matrix, and Loewy Structure," preprint, 2025. DOI: 10.5281/ZENODO.17354384.
(32) William Huanshan Chuang, "Lecture 5 – Cyclic Defect Groups and Brauer Trees," preprint, 2025. DOI: 10.5281/ZENODO.17354397.
(33) William Huanshan Chuang, "Lecture 5 – Wedderburn Decomposition and Group Algebras in Modular Settings," preprint, 2025. DOI: 10.5281/ZENODO.17360792.
(34) William Huanshan Chuang, "Lecture 6 – Block Theory, Primitive Idempotents, and Defect Groups," preprint, 2025. DOI: 10.5281/ZENODO.17360865.
(35) William Huanshan Chuang, "Lecture 6 – Defect Zero Blocks, Vertices, and Green Correspondence," preprint, 2025. DOI: 10.5281/ZENODO.17354490.
(36) William Huanshan Chuang, "Lecture 7 – Projective Modules and Projective Covers," preprint, 2025. DOI: 10.5281/ZENODO.17361304.
(37) William Huanshan Chuang, "Lecture 7 – Relative Projectivity and Induction/Restriction," preprint, 2025. DOI: 10.5281/ZENODO.17354626.
(38) William Huanshan Chuang, "Lecture 8 – Green Correspondence, Source Algebras, and Puig Theory," preprint, 2025. DOI: 10.5281/ZENODO.17354690.

(39) William Huanshan Chuang, "Lecture 8 – Using GAP for Modular Character Theory," preprint, 2025. DOI: 10.5281/ZENODO.17361349.
(40) William Huanshan Chuang, "Lecture 9 – Green Correspondence and Subgroup Relations," preprint, 2025. DOI: 10.5281/ZENODO.17361386.
(41) William Huanshan Chuang, "Lecture 9 – Rickard Equivalences and Broué's Conjecture," preprint, 2025. DOI: 10.5281/ZENODO.17354716.
(42) William Huanshan Chuang, "Lecture Notes – Modular Representation Theory and Brauer Characters of A_5," preprint, 2025. DOI: 10.5281/ZENODO.17354881.
(43) William Huanshan Chuang, "Modular Representation Theory and Physics," preprint, 2025. DOI: 10.5281/ZENODO.17361703.
(44) William Huanshan Chuang, "Modular Representation Theory in Physical Systems: A Categorical and Homological Approach to Symmetry, Duality, and Topology," preprint, 2025. DOI: 10.5281/ZENODO.17361745.
(45) William Huanshan Chuang, "Modular Representation Theory – Block Idempotents, Central Characters, and Decompositions," preprint, 2025. DOI: 10.5281/ZENODO.17362309.
(46) William Huanshan Chuang, "Modular Representation Theory – Central Characters, Tensor Products, and Decomposition," preprint, 2025. DOI: 10.5281/ZENODO.17361898.
(47) William Huanshan Chuang, "Modular Representation Theory – Notes and Expansions," preprint, 2025. DOI: 10.5281/ZENODO.17362401.
(48) William Huanshan Chuang, "Modular Representations and Algorithmic Approaches – A Journey from First Principles to Advanced Theory," preprint, 2025. DOI: 10.5281/ZENODO.17362583.
(49) William Huanshan Chuang, "Modular Representations and Brauer Characters of A_5," preprint, 2025. DOI: 10.5281/ZENODO.17354850.
(50) William Huanshan Chuang, "Modular Representations via Quotients, Tensor Products, and Projective Characters," preprint, 2025. DOI: 10.5281/ZENODO.17361591.
(51) William Huanshan Chuang, "Notes on Semisimple Algebras, Jacobson Radical, and Group Algebras," preprint, 2025. DOI: 10.5281/ZENODO.17353961.
(52) William Huanshan Chuang, "Semisimple Algebras, Minimal Ideals, and Centralizer Duality," preprint, 2025. DOI: 10.5281/ZENODO.17361652.
(53) William Huanshan Chuang, "Semisimple Modules, Jacobson Radicals, and Related Results," preprint, 2025. DOI: 10.5281/ZENODO.17353985.
(54) William Huanshan Chuang, "Basic Notes on Finite Group Representations and Total Variation Distance," preprint, 2025. DOI: 10.5281/ZENODO.17353199.
(55) William Huanshan Chuang, "Introduction to Representation Theory, Character Theory, and Applications to Random Walks," preprint, 2025. DOI: 10.5281/ZENODO.17353428.
(56) William Huanshan Chuang, "Jacobson Radical in Artinian $\mathbb{Z}$-Algebras – Nilpotency and Centrality," preprint, 2025. DOI: 10.5281/ZENODO.17352562.
(57) William Huanshan Chuang, "Lecture Notes on Random Walks and Representation Theory," preprint, 2025. DOI: 10.5281/ZENODO.17353400.
(58) William Huanshan Chuang, "Notes on Nakayama's Lemma, Jacobson Radicals, and Related Topics," preprint, 2025. DOI: 10.5281/ZENODO.17353485.

(59) William Huanshan Chuang, "Notes on Probability and Representation Theory of Finite Groups," preprint, 2025. DOI: 10.5281/ZENODO.17353286.
(60) William Chuang, "Introduction to Group Rings and Fourier Transformations in Representation Theory," preprint, 2025. DOI: 10.13140/RG.2.2.27573.95206.
(61) William Chuang, "Lecture Notes on Double Coset Random Walks," preprint, 2025. DOI: 10.13140/RG.2.2.24533.08164.
(62) William Huanshan Chuang, "Lecture Notes on the Character Table of A_5," preprint, 2025. DOI: 10.13140/RG.2.2.32607.11682.
(63) William Chuang, "Lecture on Algebras, Modules, and Representations," preprint, 2025. DOI: 10.13140/RG.2.2.29566.24642.
(64) William Chuang, "Preliminaries for Representation Theory," preprint, 2025.
DOI: 10.13140/RG.2.2.14152.17924.
(65) William Chuang, "Energy Dynamics in Self-Attention: Theoretical Insights, Counterexamples, and Alternatives to Standard Scaling Factors in Self-Attention Mechanisms," preprint, 2024.
DOI: 10.36227/techrxiv.173397945.52437090/v1.
(66) William Huanshan Chuang, "The Hausdorff Dimension of Limit Sets of Well-distributed Schottky Groups," M.A. thesis, San Francisco State University, 2022.
DOI: 10.46569/20.500.12680/v118rm53p.

National Security Relevance and Clearance Readiness

Several of the theoretical frameworks described above — including geometry-native neural architectures (GRAIL), metric-invariant arithmetic (MIA), observer-centric dual quantization (lambda-stack), post-quantum cryptographic primitives (MSIA, fractal and holographic encryption), and Langlands- and monodromy-based training schemes — have clear structural relevance to national security, cryptanalytic resilience, and post-quantum computing strategy.

A subset of this work, under the title *"Hyperbolic Autoencoder and Transformer Framework"*, was submitted as part of a formal RFI response to **DARPA-SN-25-60** and is provisionally protected under **U.S. Patent Application #63/773,441** (filed March 18, 2025). That submission specifically highlights:

- monodromy-driven hyperbolic transformer and autoencoder layers;
- Lorentzian loss-landscape modulation for global optimization;
- embedding post-quantum security features and arithmetic structure (e.g., Langlands- and automorphic-inspired mechanisms) directly into attention and encoding flows.

While some design details remain unpublished to preserve strategic and export-sensitive value, all research has been conducted in compliance with U.S. academic and regulatory standards (including ITAR/EAR public disclosure protocols). At present, all materials are unclassified and suitable for academic dissemination while remaining applicable to mission-critical domains such as secure compute, post-quantum cryptography, and robust AI deployment.

Clearance readiness. I am open to future clearance-based collaboration with U.S. national security organizations (e.g., **DoD**, **DARPA**, **NSA**, **CIA**, **FBI**), subject to standard eligibility and review procedures.

Other Independent Projects

University of Arizona

- **Independent Study: Real and Complex Analysis, and Applications of Hyperbolic Geometry**
  With Prof. David Glickenstein, Fall 2023

University of Arizona

- **RTG Project: Scaling Factors of Self-Attention Weights in Transformers**
  With Prof. Ning Hao, Fall 2023

San Francisco State University

- **Computation of the Hausdorff Dimension of Limit Sets of Schottky Groups**
  With Dr. Chun-Kit Lai, June 2021 – May 2022

San Francisco State University

- **Independent Study: Prime Geodesic Theorem and Limit Sets of Schottky Groups**
  January 2021 – May 2021
  Wrote a summary of the modern proof with an emphasis on growth rates based on the Hausdorff dimension of the associated limit set.
  Advisor: Dr. Chun-Kit Lai

San Francisco State University

- **Topology Project: A Study on Fundamental Groups**
  September 2020 – December 2020
  Advisor: Dr. Emily Clader

San Francisco State University

- **Independent Study: Hom-Polytopes**
  September 2019 – December 2019
- **Combinatorics Project: Simplicial Complexes**
  January 2019 – May 2019
  Advisor: Dr. Joseph Gubeladze

University of San Francisco

- **Independent Study: Prime Number Theorem**
  January 2018 – May 2018
  Advisor: Dr. Paul Zeitz

Pennsylvania State University–University Park

- **Functional Analysis Project: Hardy's Proof of Uniform Distribution**
  January 2018 – May 2018

- **Independent Study: Reading "Lecture Notes on Functional Analysis: With Applications to Linear Partial Differential Equations"**
  January 2018 – May 2018
  Advisors: Dr. Sergei Tabachnikov and Dr. Moisey Guysinsky

Pennsylvania State University–University Park

- **Topology Project: Solving the $(9, 8, 4, 3, 7)$-Linkage Problem**
  January 2018 – May 2018
- **Topology Final Project: Conway's Basic Theorem**
  September 2017 – December 2017
  Advisor: Dr. Sergei Tabachnikov

University of San Francisco

- **Capstone Project: Graph Theory for an Inverted-Index-Based Search Engine**
  January 2018 – May 2018
  Advisor: Dr. Chris Bryan

University of San Francisco

- **Capstone Project: Applying the Method of Steepest Descent and Cauchy Contour Integrals to the Fisher Exact Test**
  January 2018 – May 2018
  Advisor: Dr. Xuemei Chen

University of San Francisco

- **Research Assistant**
  August 2016 – May 2017
  Assisted with lecture notes for MSAN 504 (Review of Probability and Statistics).
  Advisor: Dr. Jeff Hamrick

University of San Francisco

- **Capstone Project: Implementing Dijkstra's Algorithm Applications**
  Spring 2016
- **Summer Research Project: Therapeutic Video Games for Patients with Disabilities**
  June 2016 – September 2016
- **Interpreting Deep Neural Networks**
  Fall 2016
  Explored causal structures within deep networks to map them onto symbolic graphs, and investigated methods to initialize models from human-written code.
  Read causal inference research by Prof. David Galles and Judea Pearl.
  Advisor: Dr. David Galles

Pre-Baccalaureate Independent Projects

National Taiwan University

- **Research Student at LeCosPA**
  September 2011 – May 2013
  Old Profile: Tao-Mao Chuang
  LeCosPA People (Archived)
  Presented various topics in weekly meetings and seminars, including:
  - Bremsstrahlung and Cherenkov radiation
  - Topological quantum field theory and 2+1D quantum gravity via Chern-Simons terms
  - Cosmological constant, vacuum structure, and vacuum energy
  - Radiation from moving mirrors and black holes (Schwinger mechanism, Casimir effect, Hawking/Unruh effects)
  - Potential carbon-free energy sources via low-energy nuclear reactions
  - Metamaterials and analog models of gravity
  - Instability of Anti-de Sitter space
  - Induced gravity, Coleman-Weinberg–Witten theorem on Lorentz violation, AdS/CFT correspondence
  - Quantum information, holographic turbulence, AdS/CMT, sonoluminescence
  - Holographic renormalization group flow and Ricci flow
  - Background-independent spin foam models and Regge calculus

  Advisor: Dr. Pisin Chen

National Taiwan University

- **Kontsevich–Soibelmann Wall-Crossing Formula for Mathematical QFT**
  January 2012 – May 2012
- **Conformal Bootstrap Methods for the 3D Ising Model**
  2011
  Advisor: Dr. Heng-Yu Chen

National Taiwan University

- **A Study on the Lee–Yang Theorem and Riemann Zeta Function in Statistical Mechanics**
  January 2012 – May 2012
  Advisor: Dr. Ning-Ning Pang

National Taiwan University

- **Dark Energy Problem via Modified Gravity**
  September 2010 – May 2011
  Focused on the equivalence of Einstein frames and conformal mappings in scalar-tensor theory.
  Advisor: Dr. Je-An Gu

National Taiwan University

Independent study: Reading General Relativity books and papers written by Wheeler et. al., Carroll and Dirac, Fall 2010
A Consistency Verification of Extended Theories of Gravitation, Fall 2010
Advisor: Dr. Je-An Gu

National Dong Hwa University

Research Assistant, Spring 2010
Studying Quantum Mechanics books and papers written by Griffiths, Shankar, Sakurai, and Bryon and Fuller, and solving problems on Arfken's book.
Advisor: Dr. Cheng-Pang Liu

Research Assistant, Spring 2010

Work Experience (Teaching & Research)

Logarchéon Inc. (Marana, AZ)

- Founder & Principal Researcher — June 15, 2025–Present
  Founded and operate a one-person, IP-first research C-corporation (a "geometry-native recursive agent lab") developing interpretable, certifiable transformer systems and secure, geometry-native computation frameworks. Current work includes CEAS (Critical Entropy Attention System), finite-machine lift for edit-without-retrain, GRAIL-based secure execution, the Λ-Stack, and Metric-Invariant Architecture (MIA), with applications to cryptomorphic twin deployments, invariant-first compute, and high-assurance AI for critical infrastructure.

University of Arizona

- Graduate Teaching Assistant, MATH 112 (College Algebra), Section 33 – Fall 2022
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 12 & 18 – Spring 2023
- Graduate Teaching Assistant, MATH 112 (College Algebra), Section 13 – Fall 2023
- Grader, MATH 112 (College Algebra), Sections 9, 13 & 20 – Fall 2023
- Tutor, MATH 129 (Calculus II) – Fall 2023
- Grader, MATH 129 (Calculus II) Final Exam – Fall 2023
- Grader, MATH 122B/125 (Calculus I) Common Final Exam – Fall 2023
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 101, 102, 201, 202, 401 & 402 – Spring 2024
- Graduate Teaching Assistant, MATH 125 (Calculus I), Section 001 – Fall 2024
- Graduate Teaching Assistant, MATH 112 (College Algebra), Sections 103 & 203 – Spring 2025
- **Teaching Mentors & Advisors:** Mitchell Wilson, Tina Deemer, Catherine Yslas, Oussama Ben Said, Tynan Lazarus, and Prof. David Glickenstein

San Francisco State University

- Graduate Teaching Assistant, Calculus – Spring 2022
- Grader, MATH 227 [05] (Calculus II)
- Instructor, MATH 226 [38] (Calculus I) – Fourth-hour component of MATH 226 [37]
- Instructor, MATH 227 [06] (Calculus II) – Fourth-hour component of MATH 227 [05]
- Instructor, MATH 227 [36] (Calculus II) – Fourth-hour component of MATH 227 [35]

- **Advisors:** Prof. Kim Seashore, Prof. Shandy Hauk, and Prof. Eric Hsu

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Other Academic Experience

San Francisco State University

- Graduate Teaching Assistant, Pre-Calculus – Fall 2019
- Advisor: Prof. Kim Seashore

University of San Francisco

- San Francisco Math Circle – Fall 2016
- Advisor: Prof. Paul Zeitz

National Dong Hwa University

- Undergraduate Research Assistant – Spring 2010 Hired and advised by Prof. Cheng-Pang Liu
- Tutor of Calculus and General Physics – August 2008 to December 2009 Hired by the NDHU Department of Physics

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Licenses & Certificates

- **Protecting God's Children® – Online Awareness Session 4.0**
  National Catholic Services, LLC (for Order of Malta, Western Association)
  March 7, 2025   Credential ID: V1812623
  Child-safeguarding / safe-environment training on abuse prevention, reporting obligations, and best practices for ministry with minors.
- **Safety Preparedness Training**
  The University of Arizona, Employee Development, Growth, and Engagement
  December 8, 2023
- **Information Security Awareness Certification**
  The University of Arizona, Employee Development, Growth, and Engagement
  August 27, 2023
- **President's Ambassador Medal**
  University of San Francisco
  November 2018
  Institutional medal recognizing service and representation as a USF President's Ambassador.
- **Graduate, Mathematics Advanced Study Semesters (MASS)**
  The Pennsylvania State University
  January 2018
  Completed all requirements of the 2017 Mathematics Advanced Study Semesters program in advanced undergraduate mathematics.

- **Pi Mu Epsilon National Honorary Mathematics Society**
  California Rho Chapter
  January 2017
  Elected to Pi Mu Epsilon for superior achievement in mathematics.
- **Recognition of Service Award**
  ACM SIGMOD/PODS 2016 (San Francisco)
  January 2016
  ACM SIGMOD Recognition of Service Award for contributions to the SIGMOD 2016 conference.
- **MITx — Tackling the Challenges of Big Data**
  MIT Computer Science and Artificial Intelligence Laboratory (CSAIL) / MIT Professional Education
  February 3 – March 17, 2015
  Completed a 20-hour professional course on large-scale data systems and analytics.
- **Ranked 1st of 55 — Undergraduate Physics Cohort**
  National Dong Hwa University, Department of Physics
  January 2013
  Registrar's certificate confirming class rank 1/55 (issued under the name Huan-Shan Chuang).
- **Certificate of Appreciation — Holy Week Liturgy Volunteer**
  Fu Jen Catholic University, Religious Guidance Center
  April 2004
  Recognized for assisting the 2004 Holy Week campus liturgies and helping the community "better experience the peace and joy of the Resurrection."
- **Club Officer Certificates — Science Fair Research & Computer Studies Clubs**
  St. Ignatius High School, Taipei, Taiwan
  2002–2003
  Certificates as *President*, Science Fair Research Club, and *Activities Coordinator*, Computer Studies Club (C/C++ programming for Olympiad-style problems); includes completion of the St. Ignatius Club Officers Leadership Training Camp (exemplary performance).
- **Certificate of Baptism & First Holy Communion**
  Parish of the Sacred Heart of Jesus, Shulin, Catholic Archdiocese of Taipei
  April 7, 1996
  Baptized "Thomas" and received First Holy Communion at Shulin Catholic Church; certificates list parents, godfather Der-kuang Chen, and celebrant Rev. J. Donald McGinnis.

Honors and Awards

(1) **Nominated for MSRI Summer Graduate School on Metric Geometry and Geometric Analysis**
University of Oxford (UK), Fall 2021

(2) Dean's Honor Roll Distinction — University of San Francisco (Spring 2018). Recognized on the Dean's Honor Roll for outstanding academic performance during the Spring 2018 term.

(3) **Mathematics Advanced Study Scholarship and Internal Scholarship (MASS Program)**
The Pennsylvania State University–University Park, Fall 2017
(Covered tuition and fees)

(4) **Dean's Honor Roll**
University of San Francisco, Spring 2015, Fall 2016, and Spring 2017

(5) **Pi Mu Epsilon Honor Society**
University of San Francisco

(6) **Admitted to the Summer School on Symmetry in Mathematics and Physics**
National Taiwan University, Summer 2012

(7) **Admitted to Prof. Anthony Zee's Quantum Field Theory Course**
Institute of Physics, Academia Sinica, February 2012

(8) **Admitted to the 1st LeCosPA Symposium: Towards Ultimate Understanding of the Universe**
National Taiwan University, February 2012
[Link]

(9) **Admitted to the 2nd International Workshop on Dark Matter, Dark Energy, and Matter-Antimatter Asymmetry**
National Tsing Hua University, Winter 2010
[Link]

(10) **Admitted to the Summer School for Theoretical Physics**
National Tsing Hua University, Summer 2009

(11) Presidential Semester Honor — National Dong Hwa University, Physics (Mar 2010). University presidential honor recognizing top GPA ranking in the Department of Physics (Aug 2009–Jan 2010). Conferred in person by the NDHU President. Issued under former name Tao-Mao Chuang.

(12) Presidential Semester Honor — National Dong Hwa University, Physics (Mar 2009). University presidential honor recognizing top GPA ranking in the Department of Physics for the 2008–2009 fall semester. Conferred by President Wen-Shu Hwang. Issued under former name Tao-Mao Chuang.

(13) Presidential Semester Honor — National Dong Hwa University, Physics (Nov 2008). University presidential honor recognizing top GPA ranking in the Department of Physics for the Spring 2008 term. Conferred by President Wen-Shu Hwang. Issued under former name Tao-Mao Chuang.

(14) Presidential Semester Honor — National Dong Hwa University, Physics (Mar 2008). University presidential honor recognizing top GPA ranking in the Department of Physics for the Fall 2007 term. Conferred by President Wen-Shu Hwang. Issued under former name Tao-Mao Chuang.

(15) Character and Conduct Excellence (Moral Conduct Award) — St. Ignatius High School (Jun 2006). Schoolwide recognition for exemplary conduct and citizenship. Issued under former name Tao-Mao Chuang.

(16) Taipei County Graduation Excellence Award — St. Ignatius High School (2006). County-level recognition for outstanding academic performance at high-school graduation, presented by the Taipei County Government. Included the "Zhen-Xi" Award Cup as a symbol of excellence across academics, conduct, and overall development. Issued under former name Tao-Mao Chuang.

(17) St. Ignatius Direct-Entry Scholarship — Academic Merit (Spring 2006). Merit-based Direct-Entry Scholarship awarded senior year for academic excellence in AY 2005–2006 (spring). Issued by St. Ignatius High School (Taiwan).

(18) St. Ignatius Direct-Entry Scholarship — Academic Merit (Dec 2005). Merit-based Direct-Entry Scholarship awarded senior year for academic excellence in AY 2005–2006 (fall). Issued by St. Ignatius High School (Taiwan).

(19) St. Ignatius Direct-Entry Scholarship — Academic Merit, Fall 2005. Merit-based Direct-Entry Scholarship recognizing top academic standing in the first semester of AY 2005–2006. Issued under former name Tao-Mao Chuang.

(20) Academic Excellence — First Benchmark Exam (AY 2005–2006), St. Ignatius High School. Top performance in the first semester benchmark exam of AY 2005–2006 (Grade 11). Issued under former name Tao-Mao Chuang.

(21) Arts and Humanities Excellence (Comics) — 41st Anniversary Teaching Achievement Exhibition, St. Ignatius High School (Nov 2005). Recognized for an outstanding comics piece in the school's 41st-anniversary teaching achievement exhibition (Grades 10–12; achieved in Grade 11). Issued under former name Tao-Mao Chuang.

(22) Chinese Literature Exhibit — Honorable Mention, 41st Anniversary Teaching Achievement Exhibition (Nov 2005). Honorable Mention for the essay "Constant dripping wears away stone" in the Chinese literature exhibit at the 41st-anniversary teaching achievement exhibition (Grades 10–12; achieved in Grade 11). Issued under former name Tao-Mao Chuang.

(23) Mathematics Exhibit Award — Polyhedra Model, 41st Anniversary Teaching Achievement Exhibition (Nov 2005). Recognized for an outstanding polyhedra model in the mathematics exhibit at the school's 41st-anniversary teaching achievement exhibition (Grades 10–12; achieved in Grade 11). Issued under former name Tao-Mao Chuang.

(24) English Listening Competition — Honorable Mention, St. Ignatius High School (Apr 2005). Received Honorable Mention in the Grade 11 division of the school English Listening Competition (AY 2004–2005). Issued under former name Tao-Mao Chuang.

(25) St. Ignatius Direct-Entry Scholarship — Academic Merit, Spring 2005. Merit-based Direct-Entry Scholarship for top academic standing in Spring 2005 (AY 2004–2005). Issued under former name Tao-Mao Chuang.
(26) St. Ignatius Scholarship — Academic Merit, Spring 2005. Merit-based school scholarship for top academic standing in Spring 2005 (AY 2004–2005). Issued by St. Ignatius High School (Taiwan).
(27) Academic Excellence — Review Exam, St. Ignatius High School (Mar 2005). Top performance on the semester review exam (Grade 11). Issued under former name Tao-Mao Chuang.
(28) Essay Contest (Winter Reading Program) — Honorable Mention, St. Ignatius High School (Mar 2005). Honorable Mention in the Grade 11 division of the Winter Break extracurricular reading essay competition (AY 2004–2005). Issued under former name Tao-Mao Chuang.
(29) Schoolwide Physics Competition — 1st Place Overall (Grade 10 Division), St. Ignatius High School (Feb 2005). Placed 1st in the schoolwide physics subject competition (Grade 10 division). Issued under former name Tao-Mao Chuang.
(30) Schoolwide Physics Competition — 1st Place Overall (Grades 10–12; achieved in Grade 11), St. Ignatius High School. Ranked 1st overall in the schoolwide physics competition spanning Grades 10–12 (single combined ranking). Served as the selection exam for the Physics Olympiad team; subsequently selected for Olympiad training. Earned after roughly six months of formal physics coursework.
(31) St. Ignatius Direct-Entry Scholarship — Academic Merit, Fall 2004. Merit-based Direct-Entry Scholarship awarded for top academic standing in the first semester of AY 2004–2005. Issued under former name Tao-Mao Chuang.
(32) St. Ignatius Scholarship — Academic Merit, Fall 2004. Second-class school scholarship recognizing high academic performance in the first semester of AY 2004–2005.
(33) High Distinction — First Sectional/Term Exam, AY 2004–2005, St. Ignatius High School. Top-tier award in the first sectional/term exam of AY 2004–2005 (Grade 11). Issued under former name Tao-Mao Chuang.
(34) High-School English Vocabulary Proficiency — Level 3 (Excellent), St. Ignatius High School (Oct 2004). Achieved an Excellent rating on the Level-3 English vocabulary proficiency exam in the first semester of AY 2004–2005 (Grade 11).
(35) Academic Distinction — Term Exam, St. Ignatius High School (Sep 2004). Awarded Academic Distinction for top marks on a term exam (first stage exam of the semester). Issued under former name Tao-Mao Chuang.
(36) Natural Sciences Excellence Award — Edison Commemorative Committee (2004). Recognized by the Edison Commemorative Committee for outstanding performance in mathematics and natural sciences during AY 2003–2004 among high-school students. Issued under former name Tao-Mao Chuang.
(37) School Science Fair — Physics, 1st Place (High School Division; Grades 10–12 combined), St. Ignatius High School (May 2004). Won first place in the Physics category

of the school-wide science fair (single ranking across Grades 10–12) and was selected to represent the school at the Central–North Regional Science Fair. Achieved in Grade 10.

(38) Schoolwide Physics Exam — 1st Place Overall (Grades 10–12; achieved in Grade 10), St. Ignatius High School (May 2004). Ranked 1st overall in the schoolwide physics competition spanning Grades 10–12 and was selected as the school's Physics Olympiad representative. Achieved in Grade 10, prior to any formal school physics coursework, after self-studying physics and calculus.

(39) Prose Award — Literary Arts Award, "The Wealth of Sincerity," St. Ignatius High School (May 2004). Recognized in the school's literary arts competition for the prose piece "The Wealth of Sincerity." Issued under former name Tao-Mao Chuang.

(40) Certificate of Appreciation — Holy Week Liturgy Volunteer, Fu Jen Catholic University (Apr 2004). Recognized for contributions to the 2004 Holy Week campus liturgies, assisting with ceremonies and campus ministry to help the community better experience the peace and joy of the Resurrection.

(41) Direct-Admission Scholarship — St. Ignatius High School (Apr 2004). School-issued Direct-Admission Scholarship recognizing academic performance and conduct in AY 2003–2004 (spring). Conferred with formal certificate.

(42) Outstanding Award — First Segment Exam, AY 2003–2004 Spring, St. Ignatius High School (Apr 2004). Earned an Excellent-tier award on the school-wide first segment exam of the spring term.

(43) Regional Science Fair — Physics, 1st Place (High School Division; Grades 10–12 combined), Northern Taiwan (Apr 2004). Won First Prize in Physics at the regional science fair (single combined ranking across Grades 10–12) for schools in Taipei County, Keelung County, Yilan County, and Taoyuan County; advanced from the school science fair as St. Ignatius's representative.

(44) St. Ignatius Scholarship — First-Class, Spring 2004. First-class school scholarship for top academic standing in Spring 2004 (AY 2003–2004). Issued under former name Tao-Mao Chuang.

(45) High-School English Vocabulary Proficiency — Level 2 (Excellent), St. Ignatius High School (Mar 2004). Achieved an Excellent rating on the school's English vocabulary Level-2 proficiency exam during the Spring 2004 term.

(46) Student Service Merit — St. Ignatius High School (Mar 2004). Jesuit school merit citation recognizing exemplary conduct, service, and extracurricular contribution to the school community.

(47) Schoolwide Mathematics Competition — 1st Place (Grade 10 Division), St. Ignatius High School (Dec 2003). Placed 1st in the school-wide mathematics subject competition for the Grade 10 division. Issued under former name Tao-Mao Chuang.

(48) Academic Distinction — Second Term Exam, AY 2003–2004 (Fall), St. Ignatius High School (Dec 2003). Awarded Academic Distinction for top performance on the second term exam during the first semester of AY 2003–2004.

(49) Ancient Egypt Project — Teaching Achievement Exhibition Award, St. Ignatius High School (Nov 2003). Received an excellence award for an Ancient Egypt exhibit at the

school's anniversary teaching achievement showcase, summarizing roughly five years of self-directed study in ancient civilizations.

(50) English Essay Competition — Honorable Mention (40th Anniversary), St. Ignatius High School (Nov 2003). Received Honorable Mention in the Grade 10 division of the school's 40th-anniversary English essay competition, with certificate recognition for creative writing.

(51) High-School English Vocabulary Proficiency — Level 1 (Excellent), St. Ignatius High School (Oct 2003). Earned an Excellent rating on the school's High-School English Vocabulary Level-1 proficiency exam (Grade 10).

(52) Merit Scholarship — St. Ignatius High School (Oct 2003). Awarded a merit scholarship for top academic standing in the first semester of AY 2003–2004. Issued under former name Tao-Mao Chuang.

(53) Watercolor Sketching Contest — Merit Award (Grade 9 Group), St. Ignatius High School (Oct 2003). Received a Merit Award in the Grade 9 group of the School Anniversary Watercolor Sketching Contest organized by the Arts and Skills Department.

(54) Club Officer Certificate — Activities Coordinator, Computer Science Studies Club, St. Ignatius High School (Sep 2003). Served as Activities Coordinator for the Computer Science Studies Club, commended for enthusiastic service and outstanding performance.

(55) Club Leaders Training Camp — Completion and Excellence Certificate, St. Ignatius High School (Aug 2003). Represented the Science Fair Project Study Club at the school's Club Leaders Training Camp; completed the program with an excellence rating and was cited as a role model among student officers.

(56) Science Fair Project Study Club — President, St. Ignatius High School (Jun 2003). Served as President of the Science Fair Project Study Club in Grade 9, leading science-fair activities and mentoring peers; received a Club Officer Certificate for outstanding enthusiasm and service.

(57) Special Merit Award — Truth, Reason, and Service, St. Ignatius High School (Jun 2003). Schoolwide Special Merit Award recognizing truthful conduct, reasoned thought, and dedicated service during junior high years.

(58) Taipei County Science Fair — First Prize, Junior High Division (Life and Applied Science) (Jun 2003). County-level science exhibition (Grades 7–9 combined). Project in Life and Applied Science received First Prize and was formally commended as outstanding.

(59) Chinese Language Competition — 1st Place (Grade 9 Division), St. Ignatius High School (May 2003). Placed 1st in the schoolwide Chinese language competition for the Grade 9 division.

(60) English Writing Competition — 2nd Place (Grade 9 Division), St. Ignatius High School (May 2003). Placed 2nd in the schoolwide English writing contest for the Grade 9 division.

(61) Literary Arts Award — First Prize, Essay Category (Junior High), St. Ignatius High School (May 2003). Awarded First Prize in the school Literary Arts Awards in the essay category, recognizing exceptional creative writing among the Grade 9 cohort.

(62) Physics–Chemistry Competition — 2nd Place (Grade 9 Division), St. Ignatius High School (May 2003). Placed 2nd in the schoolwide Physics–Chemistry subject competition for the Grade 9 division.

(63) School Science Fair — 1st Place, Life and Applied Sciences (Junior High Division); Taipei County Representative (May 2003). Won 1st place in the Life and Applied Sciences category at the campus science fair (Junior High division) and was selected to represent the school at the Taipei County Science Fair.

(64) Field-Trip Poster Design — Special Excellence (Off-Campus Learning, 4th Session), St. Ignatius High School (Apr 2003). Received the Special Excellence (top) award for poster design in the 4th Off-Campus Learning/Field-Trip Poster competition.

(65) Junior High English Vocabulary Proficiency — Level 5 (Distinction), St. Ignatius High School (Mar 2003). Earned Distinction on the school's Level-5 Junior High English Vocabulary Proficiency Exam.

(66) St. Ignatius Scholarship — First-Class (Xu-Hui Scholarship), St. Ignatius High School (Mar 2003). Awarded the First-Class Xu-Hui Scholarship for outstanding academic performance in the first semester of AY 2002–2003.

(67) Patriotic Education Essay Competition — 1st Place (Grade 9 Division), St. Ignatius High School (Feb 2003). First place in the schoolwide Chinese Patriotic Education essay competition for the Grade 9 division.

(68) Winter Reading Essay Contest — 1st Place (Grade 9 Division), St. Ignatius High School (Feb 2003). First place in the Grade 9 division of the Winter Break Extracurricular Reading essay competition.

(69) Physics–Chemistry Competition — 4th Place (Grade 9 Division), St. Ignatius High School (Jan 2003). Placed 4th in the schoolwide Physics–Chemistry subject competition for Grade 9, demonstrating strong improvement driven by disciplined study and efficient use of limited study hours.

(70) Science Fair Project Study Club — General Affairs Officer (Grade 9), St. Ignatius High School (Dec 2002). Served as General Affairs Officer of the Science Fair Project Study Club (junior-high division, Grade 9, first semester). Received a club officer certificate recognizing enthusiastic service and reliable support for science-fair activities.

(71) First-Class Award — "Fruit Battery" Natural Science Project, St. Ignatius High School (Nov 2002). Designed and built a "Fruit Battery" natural science project for the 39th Anniversary Teaching Achievement Exhibition; received the First-Class Award for creativity and performance.

(72) Outstanding Excellence Award — Chinese Literature Exhibit "Poetry into Painting," St. Ignatius High School (Nov 2002). Created an illustrated interpretation of classical poetry for the school's teaching-outcomes fair; received the Outstanding Excellence Award in the Chinese Literature category.

(73) Junior High English Vocabulary Proficiency Exam — Level 4 (Excellent), St. Ignatius High School (Oct 2002). Received a proficiency certificate for achieving an "Excellent" rating on the Level 4 Junior High English Vocabulary Proficiency Exam.

(74) Spring Sunshine Project Essay Competition — 3rd Place (Grade 9 Group), St. Ignatius High School (Oct 2002). Awarded 3rd place in the Spring Sunshine Project essay competition (Grade 9 division), recognizing strong performance in Chinese essay writing.

(75) Special Excellence Award — Field Trip Poster Design (Direct-Promotion Track), St. Ignatius High School (Sep 2002). Received the Special Excellence Award for poster design for the first off-campus learning trip of the direct-promotion class in the 2002 academic year.

(76) St. Ignatius Scholarship — Excellence Award, St. Ignatius High School (Sep 2002). Awarded the St. Ignatius Scholarship (Excellence Award) in Grade 8 Honors Class for the second semester of AY 2001, including a stipend in recognition of exceptional academic performance and promise.

(77) Taipei County Government Outstanding Graduate — Scholastic Excellence (Junior High) (Jul 2002). County-level recognition at junior-high graduation for academic excellence and contributions to school activities and competitions.

(78) Junior High English Vocabulary Proficiency Exam — Level 3 (Excellent), St. Ignatius High School (Jun 2002). Received a proficiency certificate for achieving an "Excellent" rating on the Level 3 Junior High English Vocabulary Proficiency Exam.

(79) Science Fair Project Study Club — Student Officer (Grade 8), St. Ignatius High School (Jun 2002). Served as a student officer of the Science Fair Project Study Club for both semesters of Grade 8 (Honors Class), receiving consecutive club officer certificates for enthusiasm and reliable service.

(80) First Place — Physics (Junior High Division), School Science Fair, St. Ignatius High School (Apr 2002). Won First Place in Physics (Junior High Division, Grades 7–9 combined) at the 2001 school science fair; selected as a representative for the Taipei County Science Exhibition and cited with a special award.

(81) Superlative Award — First Monthly Exam, Grade 8 Gifted Class, St. Ignatius High School (Apr 2002). Received the Superlative Award for top performance on the first monthly exam of the spring term in Grade 8 (Honors/Gifted class).

(82) First-Class Prize — English Exhibit "Colors," St. Ignatius High School (Jan 2002). English composition "Colors" was selected for the 39th Anniversary Teaching-Outcomes Exhibition and awarded First-Class Prize in the English category.

(83) Outstanding Excellence Award — Social Studies Exhibit "Research on the Death Penalty," St. Ignatius High School (Jan 2002). Prepared a Social Studies project examining the ethics and legality of capital punishment; received the Outstanding Excellence Award at the school's teaching-outcomes exhibition.

(84) St. Ignatius Scholarship — Encouragement Award, St. Ignatius High School (Jan 2002). Awarded the Encouragement Award tier of the St. Ignatius Scholarship in Grade 8 Honors Class for strong academic performance and potential during the first semester of AY 2001.

(85) Outstanding Excellence Award — Arts Exhibit "I Love My School," St. Ignatius High School (Nov 2001). Created an arts/creative project titled "I Love My School" for the

38th Anniversary Teaching-Outcomes Exhibition; received the top distinction in the Arts category.

(86) Outstanding Excellence Award — Social Studies Exhibit "A Journey on the Tamsui Line," St. Ignatius High School (Nov 2001). Developed a Social Studies project combining historical research and field observation along the Taipei–Tamsui railway line; received the Outstanding Excellence Award in the Social Studies category.

(87) Science Exhibit Merit Award — CSI-Style 3D Clay Reconstruction of the Peking Man Skull, St. Ignatius High School (Nov 2001). Designed and built a CSI-style 3D clay reconstruction of the Peking Man hominid skull for the Natural Science exhibit; received a Merit Award and was cited as exceptionally distinguished among peers.

(88) Art Contest — 2nd Place (Grade 8 Sketch), St. Ignatius High School (Jun 2001). Placed 2nd in the Grade 8 division of the school's art/sketching competition.

(89) Outstanding Excellence Award — Science Exhibit "Reflections on Building a Water Rocket," St. Ignatius High School (Jun 2001). Presented a natural-science exhibit documenting the design, construction, and analysis of a student-built water rocket; received the Outstanding Excellence Award at the school's teaching-outcomes exhibition.

(90) Water-Rocket Tournament — 1st Place, Distance Event, St. Ignatius High School (Jun 2001). Competed in the school's first water-rocket festival as an 8th-grade student; team earned 1st place in the long-distance event, contributing to the class's overall runner-up finish.

(91) Spring Light Project Essay Competition — 1st Place, St. Ignatius High School (May 2001). Awarded 1st place in the Spring Light Project essay competition at St. Ignatius High School; formally recognized by the principal for outstanding performance.

(92) Science Week Egg-Drop Engineering Contest — Special Award, St. Ignatius High School (Apr 2001). Designed and built an egg-protection device for the Science Week Egg-Drop Engineering Contest; successfully protected a raw egg dropped from 16 meters, earning a Special Award for outstanding performance.

(93) St. Ignatius Scholarship — Academic Merit (Second Class), St. Ignatius High School (Mar 2001). Received the school scholarship (Second Class) for top academic standing in the first semester of AY 2000–2001.

(94) Club Officer Certificate — Art & Design Officer, Science Fair Research Club, St. Ignatius High School (Feb 2001). Served as Art & Design Officer for the Science Fair Research Club in Grade 7 (Gifted Class), designing posters and visual materials for science projects; recognized with a club officer certificate for enthusiastic service and outstanding performance.

(95) English Vocabulary Proficiency Exam — Level 1 (Excellent), St. Ignatius High School (Jan 2001). Achieved an "Excellent" rating on the Junior High English Vocabulary Level-1 Proficiency Exam as a Grade 7 student.

(96) Water-Rocket Competition — Second Place Overall, St. Ignatius High School (Jan 2001). As an 8th-grade student, co-designed and launched an entry in the school's first Water Rocket Competition, earning second place overall on combined performance and results.

(97) Water-Rocket Competition — Third Place, Accuracy Event, St. Ignatius High School (Jan 2001). In the same Water Rocket Competition, earned 3rd place in the accuracy event for launch precision and target performance.

(98) Academic Excellence Award — Second Monthly Exam, Grade 7, St. Ignatius High School (Nov 2000). Recognized with a top-tier designation in the second monthly exam during the first semester of Grade 7 (Honors/Gifted Class), with a formal commendation certificate.

(99) School Science Fair — Earth Science, 2nd Place (Junior High Division), St. Ignatius High School (Nov 2000). Placed 2nd in Earth Science at the school science fair (Junior High Division, single ranking across Grades 7–9), achieved in Grade 7.

(100) Social Studies Exhibit "History Report" — Honorable Mention, St. Ignatius High School (Nov 2000). Received Honorable Mention for the Social Studies project "History Report" at the 37th Anniversary Teaching Achievement Exhibition (Junior High Division, single ranking across Grades 7–9).

(101) St. Ignatius School — Service Merit Award, St. Ignatius High School (Nov 2000). Jesuit school merit citation recognizing exemplary volunteer service at the school anniversary fair and campus activities.

(102) Outstanding Graduate (Mayor's Award) — Xinzhuang City, Taipei County (Jun 2000). City-level recognition at elementary graduation for academic excellence and contributions to school activities; presented by the Mayor of Xinzhuang City.

(103) Taipei County Government Outstanding Graduate Award — Scholastic Excellence (Elementary) (Jun 2000). County-level award at elementary graduation for academic excellence and contributions to school activities and competitions.

(104) Taipei County "Information Education Season" — Web Research Contest, First Prize (Mar 2000). Won First Prize in the elementary division of the countywide Information Education Season Web Research Contest organized by the Taipei County Government.

(105) 4th Chairman's Cup Taekwondo Championship — 3rd Place, Junior Boys Light-B (Jul 1999). Placed 3rd in the Junior Boys Light-B weight class at the 4th Chairman's Cup Taekwondo Championship organized by the Taipei County Sports Association Taekwondo Committee, representing Min'an Elementary School.

(106) Taekwondo Black Belt — Chinese Taipei Taekwondo Association (Jan 1999). Earned a black belt certified by the Chinese Taipei Taekwondo Association after several years of training at Guang Wu Dojang; competed in county-level tournaments and trained under a coach who also instructed university, police, and military units.

(107) National Abacus & Mental Arithmetic Examination — Level 6 (May 1994). Passed Level 6 of the National Abacus & Mental Arithmetic Examination administered by the Chinese National Federation of Commerce and National Abacus Association, with training emphasizing rapid calculation, working memory, and concentration.

Service to the Community

Early Service Initiatives (1998 – 2003).

- 1998 – 2003 – Altar Server (Mass Assistant) at Fu Jen Catholic University, assisting at liturgical celebrations attended by university professors and students.
- 2000 – First engagement with service learning at Boston College, accompanying my father, a seed professor for Fu Jen Catholic University's Service Learning Program. Taught elementary arithmetic to children from minority communities in Boston.

High School Service (2000 – 2006).

- 2000 – 2006 – Served a minimum of 15 hours per year at local hospitals in Taipei through St. Ignatius High School, assisting:
  - Elderly patients, feeding them spoon by spoon.
  - Children with cerebral palsy and other disabilities, ensuring they received meals with care and dignity.

University and Adult Service (2017 – Present).

- 2017 (Spring) – Teaching Assistant for Mathematics Enrichment Program at the University of San Francisco (USF), providing academic support and instructional assistance to students.
- 2023 (October 16) – 2024 (December) – Volunteer at the University of Arizona Tucson Math Circle, primarily maintaining the website to support outreach and mathematical education.
- 2025 (February 21 – Present) – Volunteer at Poverello House (Tucson, AZ), assisting the homeless community through meal service and hospitality.
- 2025 (March 29, scheduled) – Continuing volunteer service at Poverello House (Tucson).
- 2025 (May 18) – Volunteering at a Blood Drive (9 AM – 12 PM) organized by the Knights of Columbus at St. Thomas the Apostle Parish (Tucson, AZ).
- 2025 (September 19) – Volunteered at the Knights of Columbus "All-Knighter" men's social at St. Thomas the Apostle Parish (Tucson, AZ), assisting with event setup and serving food to Brother Knights and invited guests, including potential new members.
- 2025 (October 8) – Assisted with preparing and serving dinner for the Knights of Columbus Council meeting at St. Thomas the Apostle Parish (Tucson, AZ), supporting fraternal fellowship and council activities.

Future Service Commitment.

- Following my graduation from the University of Arizona (Spring 2025), I plan to increase my volunteer engagement with the Order of Malta, Poverello House, the Knights of Columbus, Catholic Action in San Diego, and other service initiatives.

Faith Formation and Study.

- Regular recipient and self-study of Opus Dei's publications, integrating its spiritual insights into daily life, professional service, and ethical decision-making.

- Active engagement with Catholic Action in San Diego, remotely supporting its evangelization efforts and operations through prayer and spiritual solidarity.
- Ongoing study of books published by the Order of Malta and its members, including the Catholic Bible, prayer books, and *Minutes with the Catechism*, deepening my understanding of faith and service.

Skills

- **Problem Solving and Adaptability:** Demonstrated ability to learn new skills quickly.
- **Programming Languages:** C/C++, Python, R, Java, Lisp, Shell Script, Sed, Awk, LaTeX, Mathematica
- **Libraries and Packages:** PyTorch, Lightning, NumPy, Pandas, scikit-learn, Matplotlib, Ogre3D
- Designing algorithms to generate examples for theoretical research in mathematics, physics, statistics, and computer science

Memberships & Affiliations

Catholic and Service-Oriented Organizations.

- **Order of Malta** — Auxiliary Member (Arizona Area), 2025–present
  Faith-rooted service in support of the sick and the poor; participation in humanitarian and community resilience initiatives as part of the Western Association.
- **Knights of Columbus** — Fourth Degree (Patriotic Degree), 2025–present
  Fraternal service through parish-based activities, charitable works, and civic ceremonies honoring Church, community, and country.

Professional and Honor Societies.

- **Phoenix Committee on Foreign Relations (PCFR)** — Member, 2025–present
  Member of a nonpartisan forum for dialogue on international affairs and global challenges, with particular interest in emerging technologies and strategic stability.
- **Phi Sigma Pi National Honor Fraternity** — Member, University of Arizona Chapter, 2025–present
  Invited to join a selective national honor fraternity that integrates scholarship, leadership, and service; membership based on meeting the organization's GPA standard and record of campus involvement.
- **Space Force Association (SFA)** — Member, 2025–present
  Civilian member with interest in space, technology, and national security; follows developments relevant to resilient and responsible use of space.
- **Pi Mu Epsilon** — Member, California Rho Chapter, 2017–present
  Lifetime membership in the National Honorary Mathematics Society, conferred at the University of San Francisco in recognition of outstanding achievement in mathematics.

Student Organizations and Early Leadership.

- **NTU Workplace Literacy & Leadership Club**, National Taiwan University — Member, 2012–2014
  Participated in a professional development society organized by NTU's College of Management, attending leadership and workplace-skills sessions with faculty and invited speakers, including future national leaders.
- **NTU Multi-lingua Conversation Club**, National Taiwan University — Member, 2012–2014
  Took part in peer-led multilingual exchange sessions to strengthen intercultural communication and global awareness.
- **Competitive Programming Club**, St. Ignatius (Middle & High School, Taiwan) — Activities Coordinator (Grade 10), then President (Grade 11), 2002–2005
  School-wide competitive programming club spanning Grades 7–12. In Grade 10, served as Activities Coordinator, organizing C/C++ problem-solving sessions and internal Olympiad-style contests; in Grade 11, was elected President and led a multi-grade team (Grades 7–12), mentoring younger students and coordinating training for informatics competitions.
- **Science Fair Research Club**, St. Ignatius (Middle & High School, Taiwan) — Member (Grade 7 onward), Officer (Grades 8–9), President (Grades 9–10), 2000–2004
  Multi-grade science research club spanning Grades 7–12. Joined as a Grade 7 member and advanced to officer, then was elected President beginning in Grade 9, leading project reviews, coordinating lab use and equipment across grade levels, mentoring younger students on research write-ups, and representing the club at school-wide science exhibitions. Received multiple club officer and completion certificates for leadership and service.